

18.5 Latin squares and incomplete blocks

A **Latin square** removes two crossed sources of nuisance variation at once. A **balanced incomplete block design** allows blocks smaller than the number of treatments, at a computable price in precision. The Latin square is orthogonal; the incomplete block design is not, and its treatment comparisons are adjusted for blocks exactly as they were adjusted for a covariate in [Section 18.1](#).

18.5.1 Latin squares

Definition 18.5.1 — Latin square design

An $a \times a$ **Latin square** is an arrangement of a symbols in an $a \times a$ array such that each symbol occurs exactly once in each row and exactly once in each column. A **Latin square design** compares a treatments on a^2 units classified by two crossed blocking factors with a levels each (rows and columns). The treatment applied to the unit in row i and column j is the symbol in cell (i, j) of a Latin square chosen at random.

The cyclic square $k(i, j) = i + j$ modulo a exists for every a ; the usual randomization permutes its rows, columns and symbols at random, which makes every unit equally likely to receive every treatment. With $k(i, j)$ the treatment in cell (i, j) , the model is additive in all three classifications:

$$y_{ij} = \mu + \rho_i + \kappa_j + \tau_{k(i,j)} + \varepsilon_{ij}, \quad \varepsilon_{ij} \stackrel{\text{iid}}{\sim} N(0, \sigma^2). \quad (18.5.1)$$

Theorem 18.5.1 — Analysis of the Latin square

Assume [Eq. 18.5.1](#) with $a \geq 3$. Write $\bar{y}_{i\cdot}$, $\bar{y}_{\cdot j}$ and $\bar{y}^{(k)}$ for the means of row i , column j and treatment k , each an average of a observations, and $\bar{y}$ for the grand mean.

- (a) The centred row, column and treatment spaces are mutually orthogonal. So **I** is the sum of five mutually orthogonal projections: onto the mean, rows, columns, treatments (ranks 1, $a - 1$, $a - 1$, $a - 1$) and error (rank $(a - 1)(a - 2)$). The corresponding sums of squares are

$$\begin{aligned} SS_R &= a \sum_i (\bar{y}_{i\cdot} - \bar{y})^2, & SS_C &= a \sum_j (\bar{y}_{\cdot j} - \bar{y})^2, \\ SS_T &= a \sum_k (\bar{y}^{(k)} - \bar{y})^2, \end{aligned}$$

$$\text{and } SSE = \sum_{i,j} (y_{ij} - \bar{y}_{i\cdot} - \bar{y}_{\cdot j} - \bar{y}^{(k(i,j))} + 2\bar{y})^2.$$

- (b) Every contrast $\sum_k c_k \tau_k$ is estimable, with least squares estimate $\sum_k c_k \bar{y}^{(k)}$, free of the row and column effects, and variance $\sigma^2 \sum_k c_k^2 / a$.

(c) The treatment F statistic satisfies

$$F_T = \frac{SS_T/(a-1)}{SSE/((a-1)(a-2))} \sim F(a-1, (a-1)(a-2), \gamma),$$

with

$$\gamma = \frac{a}{\sigma^2} \sum_k (\tau_k - \bar{\tau})^2.$$

Proof.

- (a) Take any two of the three factors, say rows and treatments. Each treatment occurs exactly once in each row, so the two-way table of counts has every entry 1, with all row and column totals a , and it satisfies the proportional frequencies condition Eq. 9.4.1. By Theorem 9.4.3 the two centred spaces are orthogonal. The same argument applies to rows and columns (each cell once) and to columns and treatments. By Theorem 6.3.7 the projection onto the model space is $\mathbf{P}_0 + (\mathbf{P}_R - \mathbf{P}_0) + (\mathbf{P}_C - \mathbf{P}_0) + (\mathbf{P}_T - \mathbf{P}_0)$, of rank $1 + 3(a-1)$. The error projection is $\mathbf{I}$ minus this, with rank $a^2 - 3a + 2 = (a-1)(a-2)$. Applying the projections to $\mathbf{y}$ gives the stated sums of squares and the residual.
- (b) Treatment k occurs once in each row and once in each column, so averaging Eq. 18.5.1 over its a cells gives $\bar{y}^{(k)} = \mu + \bar{\rho} + \bar{\kappa} + \tau_k + \bar{\varepsilon}^{(k)}$. For a contrast the constant cancels. Estimability follows as in Theorem 18.4.1(b), and by (a) and Theorem 9.4.1 the least squares estimate is the one computed from the treatment means alone. (c) is Corollary 9.6.2. ■

The price is the error: two degrees of freedom for $a = 3$ and six for $a = 4$, so small squares are often replicated. The error consists of pieces of the row-by-column, row-by-treatment and column-by-treatment interactions, which cannot be told apart (Exercise (B2)). An interaction, such as a treatment working better in some rows, distorts the treatment comparisons: the design is only as good as its additivity assumption.

Example. (Keyboard layouts, typists and sessions)

In a synthetic experiment, five typists each type a standard passage in five sessions, comparing five keyboard layouts. Typists differ greatly, and speed rises with practice. A randomized 5×5 Latin square gives each layout once to each typist and once in each session:

	S1	S2	S3	S4	S5
typist 1	2	3	1	4	5
typist 2	3	5	4	2	1
typist 3	5	1	2	3	4
typist 4	4	2	5	1	3
typist 5	1	4	3	5	2

The analysis of variance of the speeds (words per minute) is:

source	df	sum of squares	mean square	F
typists	4	689.2	172.3	82.7

sessions	4	89.8	22.44	10.8
layouts	4	73.6	18.40	8.83
error	12	24.99	2.083	
corrected total	24	877.6		

The layouts differ ($p = 0.0015$). Their means are 47.06, 50.10, 45.92, 50.30 and 48.94 words per minute, and a difference has standard error $\sqrt{2 \times 2.083/5} = 0.913$.

Had the sessions been ignored, with layouts randomized within typists, the error variance would have been about $(MS_C + (a - 1)MSE)/a$ (Exercise (B1)). Relative to the Latin square this is

$$\frac{22.44 + 4 \times 2.083}{5 \times 2.083} = 2.95.$$

Ignoring the typists instead would have cost a factor of 17.35, and no blocking at all a factor of 16.25.

```
import numpy as np
from scipy import stats

rng = np.random.default_rng(62)
a = 5
cyclic = (np.arange(a)[: , None] + np.arange(a)[None, :]) % a      # a standard square
square = cyclic[rng.permutation(a)][: , rng.permutation(a)]      # permute rows and
    columns
square = rng.permutation(a)[square]                                # and relabel the
    layouts
typist = rng.normal(0, 8, a)                                       # typists differ a lot
session = np.array([-3.0, -1.0, 0.0, 1.5, 2.5])                  # practice effect over
    sessions
layout = np.array([0.0, 2.0, -1.0, 3.0, 1.0])
y = np.round(55 + typist[: , None] + session[None, :] + layout[square]
    + rng.normal(0, 1.5, (a, a)), 1)

grand = y.mean()
row_m, col_m = y.mean(axis=1), y.mean(axis=0)
trt_m = np.array([y[square == k].mean() for k in range(a)])
ss_row = a * np.sum((row_m - grand) ** 2)
ss_col = a * np.sum((col_m - grand) ** 2)
ss_trt = a * np.sum((trt_m - grand) ** 2)
ss_tot = np.sum((y - grand) ** 2)
ss_err = ss_tot - ss_row - ss_col - ss_trt
df_err = (a - 1) * (a - 2)
ms_err = ss_err / df_err
F_trt = ss_trt / (a - 1) / ms_err
print(f"SS rows {ss_row:.1f}, columns {ss_col:.1f}, layouts {ss_trt:.1f}, error {
    ss_err:.2f} on {df_err} df")
print(f"F for layouts = {F_trt:.2f}, p = {stats.f.sf(F_trt, a - 1, df_err):.2g}")
print("layout means", trt_m.round(2), " se of a difference", np.sqrt(2 * ms_err / a)
    .round(3))
```

[Run this code online](#)

Because the sessions are ordered in time, a layout may affect speed on the next one. Such **carryover** effects are not in Eq. 18.5.1; *crossover designs*, such as Latin squares balanced for the preceding treatment, are built to estimate them.

18.5.2 Balanced incomplete block designs

Often a block cannot hold every treatment: a taster can judge only a few samples, a batch suffices for only a few runs. With blocks of size $k < t$, treatment comparisons are no longer free of block effects, and the fairest arrangement treats every pair alike.

Definition 18.5.2 — Balanced incomplete block design

A **balanced incomplete block design** (BIBD) with parameters (t, b, r, k, λ) arranges t treatments in b blocks of size $k < t$, so that each block contains k distinct treatments, each treatment appears in r blocks, and each pair of distinct treatments appears together in exactly λ blocks. Its **incidence matrix** is the $t \times b$ matrix $\mathbf{N}$ with $n_{ij} = 1$ if treatment i is in block j and 0 otherwise. Treatments are assigned to the units within each block at random, and the labels of treatments and blocks are randomized as well.

Lemma 18.5.2 — Identities for balanced incomplete block designs

For a BIBD:

- (a) $bk = tr$ and $\lambda(t - 1) = r(k - 1)$;
- (b) $\mathbf{N}\mathbf{N}^\top = (r - \lambda)\mathbf{I}_t + \lambda\mathbf{J}_t$, where $\mathbf{J}_t = \mathbf{1}\mathbf{1}^\top$;
- (c) $r > \lambda$, and $b \geq t$ (**Fisher's inequality**).

Proof.

- (a) Both sides of the first identity count the units. For the second, fix a treatment i . The r blocks containing it hold $r(k - 1)$ other units, and each of the other $t - 1$ treatments occurs in exactly λ of these blocks.
- (b) The (i, i') entry of $\mathbf{N}\mathbf{N}^\top$ counts the blocks containing both i and i' . It is r on the diagonal and λ off it.
- (c) $k < t$ gives $k - 1 < t - 1$, so $\lambda = r(k - 1)/(t - 1) < r$. The matrix in (b) has eigenvalue $r - \lambda + \lambda t$ on $\mathbf{1}$ and $r - \lambda$ on $\mathbf{1}^\perp$, all positive. So $\mathbf{N}\mathbf{N}^\top$ is nonsingular, and $t = \text{rank}(\mathbf{N}\mathbf{N}^\top) \leq \text{rank}(\mathbf{N}) \leq b$.

■

The model is the additive block model, observed only on the occupied cells:

$$y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij} \quad \text{for the } bk \text{ pairs with } n_{ij} = 1. \quad (18.5.2)$$

Let T_i be the total of the observations on treatment i , B_j the total of block j , and define the **adjusted treatment totals**

$$Q_i = T_i - \frac{1}{k} \sum_j n_{ij} B_j, \quad (18.5.3)$$

treatment i 's total minus the average level of the blocks it happened to be in. Then $\sum_i Q_i = 0$.

Theorem 18.5.3 — Intra-block analysis of a balanced incomplete block design

Assume Eq. 18.5.2 for a BIBD, with $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$.

- (a) $\sum_i c_i \tau_i$ is estimable iff $\sum_i c_i = 0$. Its least squares estimate and variance are

$$\sum_i c_i \hat{\tau}_i = \frac{k}{\lambda t} \sum_i c_i Q_i, \quad \text{Var}\left(\sum_i c_i \hat{\tau}_i\right) = \frac{k}{\lambda t} \sigma^2 \sum_i c_i^2. \quad (18.5.4)$$

- (b) The sum of squares for treatments adjusted for blocks is $\text{SS}_{T|B} = \frac{k}{\lambda t} \sum_i Q_i^2$ on $t-1$ degrees of freedom. The residual sum of squares is $\text{SSE} = \text{SS}_{\text{tot}} - \text{SS}_B - \text{SS}_{T|B}$, where $\text{SS}_B = \sum_j B_j^2/k - (\sum_j B_j)^2/(bk)$ is the unadjusted block sum of squares, on $bk - b - t + 1$ degrees of freedom.
- (c) Under normality,

$$F = \frac{\text{SS}_{T|B}/(t-1)}{\text{SSE}/(bk - b - t + 1)} \sim F(t-1, bk - b - t + 1, \gamma),$$

with

$$\gamma = \frac{\lambda t}{k\sigma^2} \sum_i (\tau_i - \bar{\tau})^2.$$

- (d) A randomized complete block design with r blocks estimates each contrast with variance $\sigma^2 \sum_i c_i^2/r$. The ratio of that variance to Eq. 18.5.4 is the **efficiency factor**

$$E = \frac{\lambda t}{rk} = \frac{t(k-1)}{(t-1)k} < 1.$$

Proof. Let $\mathbf{X}_B$ be the $n \times b$ indicator matrix of blocks (with the intercept in its span) and $\mathbf{Z}$ the $n \times t$ indicator matrix of treatments, with $n = bk$. Let $\mathbf{M}_B$ project onto $\mathcal{C}(\mathbf{X}_B)$, which replaces every observation by its block mean, and put $\tilde{\mathbf{Z}} = (\mathbf{I} - \mathbf{M}_B)\mathbf{Z}$. By Lemma 9.2.1, $\mathcal{C}([\mathbf{X}_B, \mathbf{Z}]) = \mathcal{C}(\mathbf{X}_B) \oplus \mathcal{C}(\tilde{\mathbf{Z}})$, orthogonally. This is the structure of Theorem 18.1.1 with the treatment indicators in the role of covariates, except that $\tilde{\mathbf{Z}}$ is not of full rank: $\mathbf{Z}\mathbf{1} = \mathbf{1}$ lies in $\mathcal{C}(\mathbf{X}_B)$, so $\tilde{\mathbf{Z}}\mathbf{1} = \mathbf{0}$.

The matrix $\mathbf{C} = \tilde{\mathbf{Z}}^\top \tilde{\mathbf{Z}} = \mathbf{Z}^\top \mathbf{Z} - \mathbf{Z}^\top \mathbf{M}_B \mathbf{Z}$ is the *information matrix* of the design. Here $\mathbf{Z}^\top \mathbf{Z} = r\mathbf{I}$, and $\mathbf{Z}^\top \mathbf{M}_B \mathbf{Z} = \mathbf{N}\mathbf{N}^\top/k$, because $\mathbf{Z}^\top \mathbf{X}_B = \mathbf{N}$ and $\mathbf{X}_B^\top \mathbf{X}_B = k\mathbf{I}$. By Lemma 18.5.2(b) and (a),

$$\mathbf{C} = r\mathbf{I} - \frac{(r-\lambda)\mathbf{I} + \lambda\mathbf{J}}{k} = \frac{r(k-1) + \lambda}{k}\mathbf{I} - \frac{\lambda}{k}\mathbf{J} = \frac{\lambda t}{k}\left(\mathbf{I} - \frac{1}{t}\mathbf{J}\right).$$

Also $\tilde{\mathbf{Z}}^\top \mathbf{y} = \mathbf{Z}^\top \mathbf{y} - \mathbf{Z}^\top \mathbf{M}_B \mathbf{y} = \mathbf{Q}$, since $\mathbf{Z}^\top \mathbf{y}$ is the vector of treatment totals and $\mathbf{Z}^\top \mathbf{M}_B \mathbf{y}$ has entries $\sum_j n_{ij} B_j/k$.

- (a) A vector $(\mathbf{0}, \mathbf{c})$, zero on the block and intercept parameters, is estimable iff $\mathbf{c}^\top = \mathbf{a}^\top \mathbf{Z}$ for some $\mathbf{a}$ with $\mathbf{a}^\top \mathbf{X}_B = \mathbf{0}$ (Theorem 8.2.2). Such an $\mathbf{a}$ satisfies $\mathbf{a} = (\mathbf{I} - \mathbf{M}_B)\mathbf{a}$, so $\mathbf{c} \in \mathcal{C}(\tilde{\mathbf{Z}}^\top) = \mathcal{C}(\mathbf{C})$, which is the space of contrasts because $\mathbf{C}$ is a positive multiple of the centring projection. Conversely, for a contrast $\mathbf{c} = \mathbf{C}\mathbf{h}$, the vector $\mathbf{a} = \tilde{\mathbf{Z}}\mathbf{h}$ works. Its least squares estimate is $\mathbf{a}^\top \mathbf{y} = \mathbf{h}^\top \mathbf{Q}$. With the generalized

inverse $\mathbf{C}^- = \frac{k}{\lambda t} \mathbf{I}$, which satisfies $\mathbf{C}\mathbf{C}^- \mathbf{C} = \mathbf{C}$ because $\mathbf{I} - \mathbf{J}/t$ is idempotent, we have $\mathbf{C}\mathbf{C}^- \mathbf{c} = \mathbf{c}$ for a contrast, so we can take $\mathbf{h} = \mathbf{C}^- \mathbf{c}$, and the estimate is $\frac{k}{\lambda t} \mathbf{c}^\top \mathbf{Q}$. Its variance is $\sigma^2 \|\tilde{\mathbf{Z}}\mathbf{h}\|^2 = \sigma^2 \mathbf{h}^\top \mathbf{C}\mathbf{h} = \sigma^2 \mathbf{c}^\top \mathbf{C}^- \mathbf{c} = \frac{k}{\lambda t} \sigma^2 \|\mathbf{c}\|^2$.

- (b) The extra sum of squares for treatments after blocks is the squared length of the projection onto $\mathcal{C}(\tilde{\mathbf{Z}})$, $\mathbf{y}^\top \tilde{\mathbf{Z}}\mathbf{C}^- \tilde{\mathbf{Z}}^\top \mathbf{y} = \mathbf{Q}^\top \mathbf{C}^- \mathbf{Q} = \frac{k}{\lambda t} \sum_i Q_i^2$ (Lemma 9.2.1). The block sum of squares is $\|(\mathbf{M}_B - \mathbf{P}_0)\mathbf{y}\|^2$, and the block, adjusted treatment and residual projections are mutually orthogonal and sum to $\mathbf{I} - \mathbf{P}_0$ (Theorem 6.3.7). The model has rank $b + t - 1$.
- (c) is Theorem 11.1.1. The noncentrality is the same quadratic form evaluated at the mean, where $\tilde{\mathbf{Z}}^\top \boldsymbol{\mu} = \mathbf{C}\boldsymbol{\tau}$, giving $\boldsymbol{\tau}^\top \mathbf{C}\mathbf{C}^- \mathbf{C}\boldsymbol{\tau} / \sigma^2 = \frac{\lambda t}{k} \sum_i (\tau_i - \bar{\tau})^2 / \sigma^2$.
- (d) The ratio is $(1/r)/(k/(\lambda t))$, and $\lambda = r(k-1)/(t-1)$ by Lemma 18.5.2(a). Since $k < t$, $(k-1)/k < (t-1)/t$.

■

The adjusted total Q_i compares treatment i with the blocks it appeared in, and $E Q_i = \frac{\lambda t}{k} (\tau_i - \bar{\tau})$ with no β 's left (Exercise B3): this is the **intra-block analysis**. The efficiency factor is the fraction of information that survives the elimination of blocks: about one half for $k = 2$, close to one for $k = t - 1$. Smaller blocks are often more homogeneous, and the smaller σ^2 can more than make up for E . Every pair sharing a block makes the design connected (Theorem 8.6.1), and balance gives every $\tau_i - \tau_{i'}$ the same variance $2k\sigma^2/(\lambda t)$. Unbalanced connected designs are analysed the same way with a different information matrix $\mathbf{C}$.

Example. (Seven recipes, fourteen tasters)

In a synthetic tasting experiment, seven recipes of a sports drink are rated on a 0–10 scale by 14 tasters, each of whom can judge only three. The blocks are the lines of two Fano planes, the cyclic sets $\{0, 1, 3\} + s$ and $\{0, 1, 5\} + s$ modulo 7 for $s = 0, \dots, 6$ (recipes numbered $0, \dots, 6$ in this description and $1, \dots, 7$ below). This is a BIBD with $t = 7$, $b = 14$, $k = 3$, $r = 6$ and $\lambda = 2$. There are 42 ratings and 22 error degrees of freedom.

Tasters use the scale very differently: the unadjusted block sum of squares is 69.72 out of a corrected total of 80.71. The raw recipe means, 6.15, 6.10, 5.40, 6.60, 6.30, 6.85 and 6.02, therefore mix recipe quality with the generosity of the tasters each recipe met. The adjusted totals are $Q = (-1.30, 0.90, -3.43, 3.13, 0.43, -1.50, 1.77)$, and the adjusted means $\bar{y} + \frac{k}{\lambda t} Q_i$ are 5.92, 6.40, 5.47, 6.87, 6.30, 5.88 and 6.58. Recipe 6 had the highest raw mean. Its adjusted mean is

$$6.202 + \frac{3}{14} \times (-1.50) = 5.88,$$

among the lowest: recipe 6 happened to meet generous tasters, and the adjustment removes 0.97 from its mean.

The adjusted recipe sum of squares is 6.36, and the residual sum of squares is 4.64, so

$$F = \frac{6.36/6}{4.64/22} = 5.03$$

on 6 and 22 degrees of freedom ($p = 0.0022$). A difference of two recipes has standard error $\sqrt{2 \times 3 \times 0.2108/14} = 0.301$. A complete block design with six tasters tasting all seven recipes, with the same σ^2 , would give 0.265; the variance ratio is the efficiency

factor $7 \times 2 / (6 \times 3) = 0.7778$. A taster judging seven drinks would probably not rate as consistently, though.

```
import numpy as np
from scipy import stats

t, k = 7, 3
blocks = [sorted((d + s) % t for d in base) for base in ([0, 1, 3], [0, 1, 5]) for s
            in range(t)]
b = len(blocks)                                # 14 tasters
N = np.zeros((t, b), dtype=int)                # incidence: recipe i tasted
                                           by taster j
for j, blk in enumerate(blocks):
    N[blk, j] = 1
r = N.sum(axis=1)[0]
lam = (N @ N.T)[0, 1]

rng = np.random.default_rng(77)
tau = np.array([0.0, 0.4, -0.3, 0.8, 0.1, -0.6, 0.5]) # recipe effects
taster = rng.normal(0, 1.2, b)                    # tasters use the scale
                                           differently
rows = [(i, j) for j, blk in enumerate(blocks) for i in blk]
recipe = np.array([i for i, j in rows])
block = np.array([j for i, j in rows])
y = np.round(6 + tau[recipe] + taster[block] + rng.normal(0, 0.5, len(rows)), 1)
n = len(y)

T_tot = np.bincount(recipe, weights=y, minlength=t) # recipe totals
B_tot = np.bincount(block, weights=y, minlength=b) # taster totals
Q = T_tot - N @ B_tot / k                        # adjusted recipe
                                           totals
tau_hat = k * Q / (lam * t)                      # intra-block estimates
                                           (sum to zero)
ss_trt_adj = k / (lam * t) * np.sum(Q ** 2)
ss_blk = np.sum(B_tot ** 2) / k - y.sum() ** 2 / n # blocks, ignoring
                                           recipes
ss_tot = np.sum((y - y.mean()) ** 2)
ss_err = ss_tot - ss_blk - ss_trt_adj
df_err = n - b - t + 1
ms_err = ss_err / df_err
F = ss_trt_adj / (t - 1) / ms_err
se_diff = np.sqrt(2 * k * ms_err / (lam * t))
eff = lam * t / (r * k)                          # efficiency factor
print("raw recipe means      ", (T_tot / r).round(2))
print("adjusted recipe means", (y.mean() + tau_hat).round(2))
print(f"adjusted recipe SS {ss_trt_adj:.3f}, error SS {ss_err:.3f} on {df_err} df,"
      f" F = {F:.2f}, p = {stats.f.sf(F, t - 1, df_err):.2g}")
print(f"se of a difference {se_diff:.3f}; efficiency factor {eff:.4f}")
```

[Run this code online](#)

Remark (Recovering inter-block information). The block totals, with $E B_j = k\mu + \sum_i n_{ij}\tau_i + k\beta_j$, also carry treatment information. With random block effects they give an independent *inter-block* estimate of every contrast, whose variance grows with σ_β^2 . Combining the two estimates by inverse-variance weights, as Yates proposed, is gener-

alized least squares in a mixed model (Chapter 32). With blocks as different as these tasters, the inter-block estimate adds little.

18.5.3 Exercises

18.5.3.1 A. Check your understanding

Exercise A1.

Show that the cyclic array $k(i, j) = i + j \bmod a$ is a Latin square, and that permuting its rows, its columns, or its symbols gives a Latin square. For $a = 3$, how many distinct Latin squares are there, and does the randomization described above reach all of them?

Exercise A2.

Check the identities of Lemma 18.5.2 for the design of Example (Seven recipes, fourteen tasters). Is there a BIBD with $t = 7$, $k = 3$ and $b = 7$? With $t = 6$, $k = 3$, $b = 6$?

Exercise A3.

From the numbers in Example (Seven recipes, fourteen tasters), compute the estimate of $\tau_4 - \tau_3$ and its standard error, and decide whether recipe 4 is better than recipe 3 at level 0.05, considered alone.

18.5.3.2 B. Practice

Exercise B1.

In a Latin square, suppose the columns had not been used as a blocking factor, so that the treatments were randomized within rows only. Following Proposition 18.4.4, show that the expected error mean square of that design is $\sigma^2 + \sum_j (\kappa_j - \bar{\kappa})^2 / (a - 1)$, and that $(MS_C + (a - 1)MSE)/a$ is unbiased for it in the Latin square.

Solution. Randomizing within rows, the units of row i are assigned to the treatments by a random permutation. Within a row, the column effects $\kappa_1, \dots, \kappa_a$ play the role of the unit values, with within-row variance $S_\kappa^2 = \sum_j (\kappa_j - \bar{\kappa})^2 / (a - 1)$, the same for every row. By Proposition 18.4.2(b), applied to the unit values $\kappa_j + \varepsilon_{ij}$ and averaged over the errors, the expected error mean square of the row-blocked design is $\sigma^2 + S_\kappa^2$. In the Latin square, $EMS_C = \sigma^2 + aS_\kappa^2$ and $EMSE = \sigma^2$ (Theorem 9.6.1), so $E\{MS_C + (a - 1)MSE\}/a = \sigma^2 + S_\kappa^2$.

Exercise B2.

In a 4×4 Latin square, the row-by-column interaction space of the 16 cells has dimension 9. Show that the treatment space (centred) lies inside it, and that the error space is

its orthogonal complement within it, of dimension 6. What does this imply about the effect of a row-by-column interaction on the treatment comparisons?

Solution. The row-by-column interaction space is the orthogonal complement, within $\mathbb{R}^{16}$, of the span of the mean, row and column indicators; it has dimension $16 - 1 - 3 - 3 = 9$. By [Theorem 18.5.1\(a\)](#) the centred treatment space is orthogonal to the mean, row and column spaces, so it lies in this complement, with dimension 3. The error space is the orthogonal complement of the mean, row, column and treatment spaces, so it is the rest of the interaction space, of dimension 6. A row-by-column interaction in the mean vector is a vector of the 9-dimensional space, and its component in the treatment space is indistinguishable from a treatment effect. Unless the interaction happens to be orthogonal to the treatment space, it biases the treatment comparisons.

Exercise B3.

Show that $E Q_i = \frac{\lambda t}{k}(\tau_i - \bar{\tau})$ in a BIBD, directly from [Eq. 18.5.2](#) and [Lemma 18.5.2](#).

Solution. $E T_i = r\mu + r\tau_i + \sum_j n_{ij}\beta_j$ and $E B_j = k\mu + \sum_{i'} n_{i'j}\tau_{i'} + k\beta_j$. So

$$E Q_i = r\tau_i - \frac{1}{k} \sum_j \sum_{i'} n_{ij} n_{i'j} \tau_{i'} = r\tau_i - \frac{1}{k} (r\tau_i + \lambda \sum_{i' \neq i} \tau_{i'}),$$

the μ 's and β 's cancelling. Writing $\sum_{i' \neq i} \tau_{i'} = t\bar{\tau} - \tau_i$ gives $E Q_i = \tau_i(r - (r - \lambda)/k) - \lambda t\bar{\tau}/k = \frac{\lambda t}{k}(\tau_i - \bar{\tau})$, using $r(k - 1) + \lambda = \lambda t$.

Exercise B4.

A **Graeco-Latin square** superimposes two Latin squares of the same order so that every ordered pair of symbols occurs exactly once. Write down one of order 3, state the model with rows, columns and two treatment factors, and give the analysis of variance table with its degrees of freedom. How many error degrees of freedom remain for $a = 3$, 4 and 5?

18.5.3.3 C. Going deeper

Exercise C1.

For a general incomplete block design with incidence matrix $\mathbf{N}$, replication r_i and block sizes k_j , show that the information matrix is $\mathbf{C} = \text{diag}(r_i) - \mathbf{N} \text{diag}(1/k_j) \mathbf{N}^\top$, that $\mathbf{C}\mathbf{1} = \mathbf{0}$, and that every treatment contrast is estimable iff $\text{rank}(\mathbf{C}) = t - 1$. Relate this to the connectedness of [Theorem 8.6.1](#).